

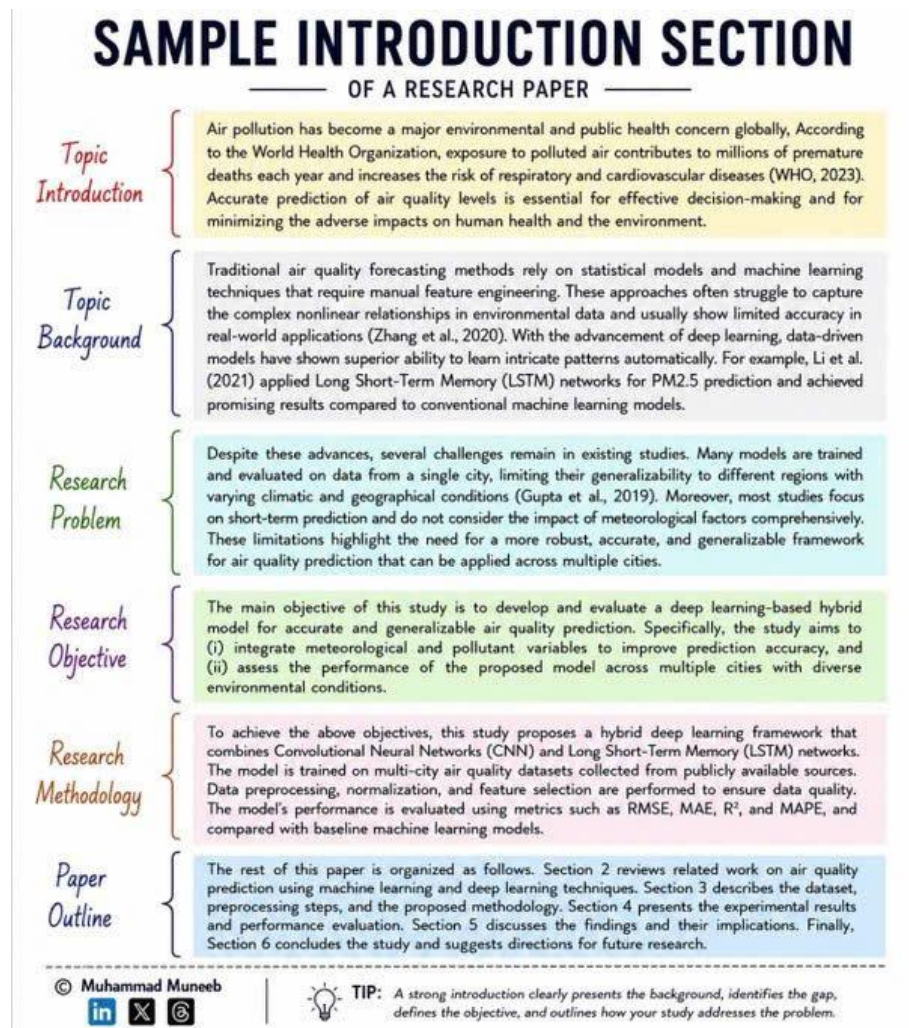
Title:

Cần nhắc tên: *CTU-Chat: A Supervisor-Routed Multi-Agent System for Heterogeneous University Counseling*

Abstract:

- Dài: nên là khoảng 150 – 300 từ
- Còn nặng về kỹ thuật và số liệu: quá chi tiết.
- Thiếu phần kết luận conclusion

Introduction (nên viết theo cấu trúc trong ảnh)



- Bối cảnh: có thể cần nhắc bổ sung thêm câu về các hệ thống chatbot/AI counseling hiện có tại các đại học khác.
- Viết ngắn gọn, cô đọng lại vì có sự lặp: phần đóng góp chính (C1, C2, C3) gần như lặp lại hoàn toàn nội dung của 3 gaps/RQs.

- Rà soát lại từ ngữ, tránh dùng khẳng định mạnh nhưng thiếu trích dẫn 😊: ví dụ “Monolithic RAG pipelines risk a uniform representation fallacy”. Các em có thể dùng từ “fallacy” nếu tìm được tài liệu tham khảo.
- Bổ sung thêm phần paper outline thành đoạn văn cuối cùng.

Conclusion

- Tránh viết giống abstract
- Bổ sung thêm về future work
- Tham khảo

Abstract vs Conclusion in a Research Paper

How to write each section, common mistakes, and what makes them different

ABSTRACT

- Purpose**

A brief summary of the entire paper. It tells the reader what was studied, how it was studied, the main findings, and why it matters.
- Writing Process**
 - Start with the topic or problem
 - State the objective or research question
 - Mention the method briefly
 - Present the key result
 - End with the main implication
- Example Abstract**

Rock slope instability remains a major challenge in mountainous road projects.	Topic / Problem
This study investigated the effect of discontinuity orientation on slope stability using field mapping and numerical modelling.	Objective + Method
The results showed that adverse joint orientation reduced the factor of safety significantly.	Key Result
The findings highlight the need for orientation-based support design in rock slopes.	Implication
- Mistakes to Avoid**
 - Too much background
 - No clear result
 - Adding citations or detailed discussion
 - Writing vague statements like 'results are discussed'

CONCLUSION

- Purpose**

The final takeaway of the paper. It explains what the findings mean, why they matter, and what should happen next.
- Writing Process**
 - Restate the aim of the study
 - Summarize the main findings
 - Interpret their significance
 - State practical implications or recommendations
 - End with a strong closing insight or future direction
- Example Conclusion**

This study examined how discontinuity orientation influences rock slope stability.	Restate Aim
The analysis confirmed that unfavorable joint sets can markedly reduce stability and increase support demand.	Main Findings
These findings show that slope design should account for structural geology from the earliest investigation stage.	Meaning / Significance
Future work should validate the results using site monitoring and case histories.	Future Work / Recommendation
- Mistakes to Avoid**
 - Repeating the abstract word for word
 - Introducing new data
 - Making claims not supported by results
 - Ending without a clear takeaway

Abstract Vs. Conclusion

Aspect	Abstract	Conclusion
Location	Beginning of the study	End of the study
Main purpose	Gives a quick overview of the entire study	Provides the final answer and meaning of the study
Answers	<i>What did you study, how, and what did you find?</i>	<i>What do the findings mean, and what should we take away?</i>
Content  Peter Munene	Problem, purpose, methodology, key findings, conclusion	Key findings, interpretation, contribution, implications, limitations, recommendations
Length	Usually brief: ~150–300 words	Usually much longer
Detail	Highly condensed	More developed and reflective
Literature citations	Usually avoided	May be used where appropriate
Research questions	Briefly indicates whether/how they were answered	Explicitly demonstrates how they were answered
Recommendations	Usually not included	Often included
Contribution	Briefly stated	Explained more fully
Reader's goal	Decide whether the study is relevant	Understand what the study ultimately means

Related work

Cần nhắc bổ sung thêm tài liệu tham khảo liên quan đến:

- Vietnamese NLP (PhoBERT, VietAI)
- Dialogue systems / intent classification
- Knowledge graph QA
- Educational AI systems ngoài REBot

Proposed Model (chỉ tập trung vào mô hình đề xuất)

- Mục 3.1: cân nhắc để tổng hợp với đoạn 2 trong Mục “Introduction” để viết cô đọng thành 1 đoạn văn trong “Introduction”.
- Viết cô đọng hơn (hiện giờ là 8 trang trong khi tổng số trang của bài báo dài là 12-16 trang)
- Rà soát lại các bảng để tránh lặp lại những từ dư thừa. Ví dụ:

Specialist Agent	Shared State	Available Direct Tools	Primary Evidence Store	Retrieval Strategy	Tool-Call Bound
Academic Specialist	Redis S, H	6 Cypher tools (trả_cuu_sganh, etc.)	Neo4j Knowledge Graph	Direct Cypher topological traversal	$\text{max_tool_calls} = 1$
Financial Specialist	Redis S, H	4 tools (tinh_toan_hoc_phi, etc.)	Tuition Graph + JSON Catalog	Pre-retrieved Fee Graph/JSON + Calc	$\text{max_tool_calls} = 1$
Scholarship Specialist	Redis S, H	1 tool (tinh_tien_hoc_bong)	Hybrid RAG + Criteria Catalog	Pre-retrieved Hybrid + Criteria tool	$\text{max_tool_calls} = 1$
General Specialist	Redis S, H	0 direct tools (pure synthesis)	PostgreSQL + Qdrant	Parent-child Hybrid RRF + BGE	$\text{max_tool_calls} = 0$

- Cần nêu rõ:
 - o CTU-Chat khác gì so với một LangGraph StateGraph với conditional edges?
 - o Đóng góp algorithm mới là gì?
 - o Mô tả vắn tắt về Neo4j Knowledge Graph schema: Node types? relationship types? Properties? Bao nhiêu nodes/edges? 6 Cypher tools cụ thể thực hiện queries gì?

Experiments

Bài chỉ so sánh nội bộ (T1–T7 ablation). Không có **bất kỳ baseline bên ngoài nào**.

Cân nhắc bổ sung thêm baseline

- GPT-4/Gemini vanilla (zero-shot hoặc few-shot)
- Single-agent RAG (LangChain standard pipeline)
- REBot [11] — được cite nhưng không benchmark
- Adaptive-RAG [8] — cũng cite nhưng không benchmark

Discussion

Chỉ tóm tắt kết quả, không phân tích sâu. Nên trả lời câu hỏi như:

- Put your results in perspective with other reports
- Explain significance of results
- What are the novel findings?
- Case studies cho thấy khi nào graph path tốt hơn và khi nào hybrid RAG tốt hơn